Time to Longitude

Converting Hours, Minutes and Seconds to Degrees

Why time becomes angle

Earth turns once relative to the mean Sun in twenty-four hours of solar time. That full turn is three hundred sixty degrees of longitude. Divide evenly and each hour of solar-time difference corresponds to fifteen degrees of longitude: two hours, thirty degrees. We work with solar time here, not with the slightly shorter sidereal day of a pure spin against the stars.

Keep the two unit systems apart. Time uses hours, minutes, and seconds (h, m, s). Angle uses degrees, arcminutes, and arcseconds (° , ′ , ″). One degree equals sixty arcminutes; one arcminute equals sixty arcseconds. One hour of time equals sixty minutes; one minute equals sixty seconds. The bridge: one hour of time = 15° of angle; one minute of time = 15′ of arc; one second of time = 15″ of arc.

The formula

Longitude difference in degrees = 15 × (hours + minutes/60 + seconds/3600)

Worked example: 2 h 35 m west

Suppose local solar time is two hours thirty-five minutes behind the time at Greenwich, compared at the same instant. Work row by row.

- 2 h of time → 2 × 15° = 30°
- 35 m of time → 35 × 15′ = 525′ of arc. Convert: 525 ÷ 60 = 8° with remainder 45′, so 8°45′.
- Add: 30° + 8°45′ = 38°45′00″ west of Greenwich.
- As a decimal: 38.75° W. Signed convention (east positive, west negative): −38.75°.

Which side of Greenwich?

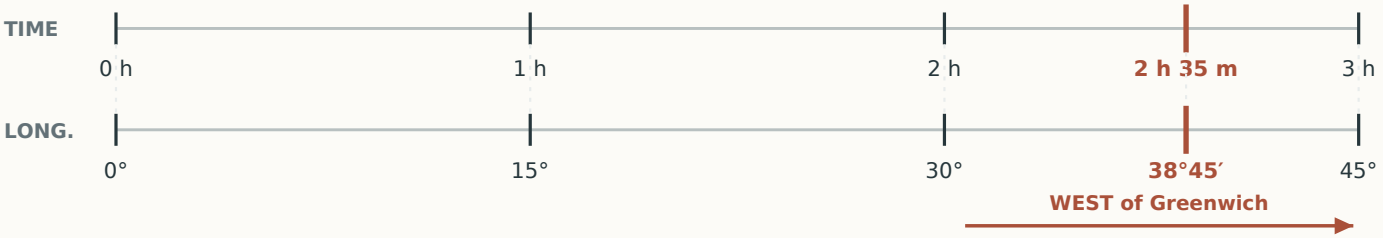
If local solar time is behind Greenwich at the same instant, you are west. If local solar time is ahead, you are east. Picture Greenwich at 12:00 while a western place still shows 9:25: that is 2 h 35 m behind, hence 38°45′ west. Always name the reference meridian—today Greenwich is usual; Chappe's published result, for example, used Paris.

This is a solar-time conversion, not a way to read exact longitude from today's time-zone label. Civil and political zones, and daylight-saving shifts, are administrative clocks, not precise meridians.

Coordinates may be written both ways. Computers commonly work with decimal degrees; people often read degrees, minutes and seconds. GPS and mapping apps can display either.

Paired scales — solar-time difference ↔ longitude west of Greenwich

West increases to the right. Endpoint at 2 h 35 m = 38°45′ (38.75°) in true proportion.



Recall — If local solar time is one hour behind Greenwich, what longitude does that suggest? Fifteen degrees west of Greenwich.